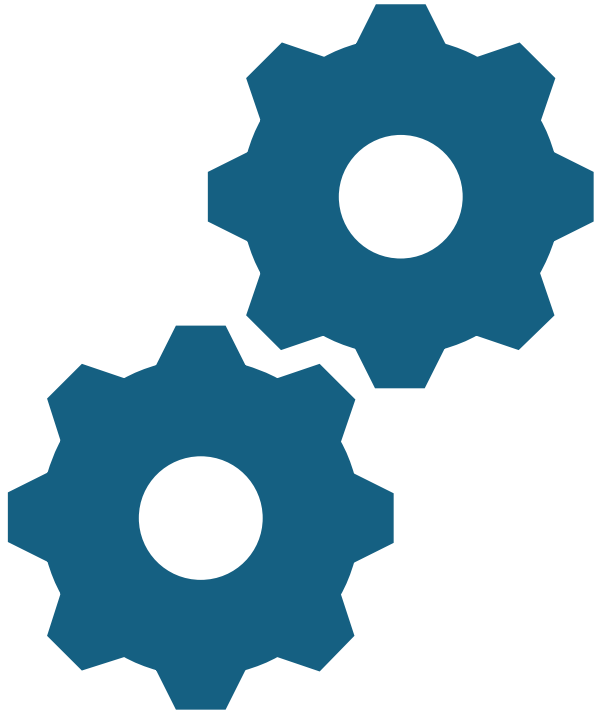
Advanced Deterministic Systems Optimization



ISE 555 Unit 5: Revised and Dual Simplex Method

Overview

- Part 1: Recap
- Part 2: Two-phase Method of Artificial Variables
- Part 3: Big M Method
- Part 4: Dual Simplex

Part 1:Recap

•Re-cap

- Last class we looked at the Simplex Method for solving Linear Programs
- Talked about the Simplex Method strategy and showed how it works – introduced Tableaus as a way to manage all of the linear algebra
- This week we will continue with the Simplex Method – introducing revised (Big-M) method and the dual simplex
- Note: This topic is also covered in Chapter 4 and Chapter 5 if you are using the textbook

Review of the Simplex Method approach

We know that for a Linear Program that is in Standard Form, the optimal solution can only occur at one of the corner points of the feasible region – these points correspond to ***basic feasible solutions*** of the constraints.

1) Start with a problem in standard form
2) Find an initial basic feasible solution:
   - If we can find a basis that leads to feasible that's great
   - What makes it a basis?
   - What can we do if that fails?

NEW FOR TODAY

   - Two-phase approach first solves a modified problem with artificial variables a_i and the objective $\min \sigma\ a_i$ The first phase moves from infeasible to feasible objectives by reducing the a_i values The second phase is the "regular simplex"
   - The Big-M method simplifies the two-phase approach by adding $\sigma\ Ma_i$ to the existing objective Then both the movement to the feasible and "regular simplex" are done simultaneously

3) We can use Tableaus to move from one basic feasible solution to a better one
4) We have a stopping criteria when all of the components of the objective row in the Tableau are non-negative

- Issues with this approach?
  - What happens when solution goes to infinity?
  - Between standard form and artificial variables, the problem size can grow significantly

What happens when we can't see a good choice for an initial basic solution?

- The Simplex Tableau examples we used the slack variables as a basis – this works when every constraint has a slack variable (they don't appear in the objective). What do we do when this isn't the case?
- We could try going through every potential basis (i.e. combinations of variables) and looking for a valid one (remember not all will lead to basic feasible solutions), but that can get tedious. For any that are in the objective, we must use the constraints to rewrite the objective without them before we start.
- We get around this issue by adding in ***artificial variables***
- ***The idea is to add extra variables to every constraint that doesn't have a slack variable. Then we solve an auxiliary optimization problem to minimize the sum of these added variables. This solution will be a basic feasible solution for the original problem – then you can start the regular optimization.***

- Why should this work?

Part 2:Artificial Variables

WHITE BOARD EXAMPLES

Two-phase Method of Artificial Variables

- Approach is simple: We add an artificial variable to every constraint without a slack variable – then use the slack + artificial variables as a basis for ***Phase 1*** of this two phase approach.

- Example:

$$\begin{aligned} \text{maximize} \quad & z = 2x_1 + x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 10 \\ & -x_1 + x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

- Standard form:

$$\begin{aligned} \text{minimize} \quad & z = -2x_1 - x_2 \\ \text{subject to} \quad & x_1 + x_2 + s_1 = 10 \\ & -x_1 + x_2 - e_2 = 2 \\ & x_1, x_2, s_1, e_2 \geq 0 \end{aligned}$$

- Why not just use (s_1, e_2) as the basic variables for a basis?

  That would not meet constraints, as $(x_1, x_2, s_1, e_2) = (0,0, s_1, e_2)$ leads to $s_1 = 10, e_2 = -2$

  Look on the graph to see why this is not a feasible solution.

Two-phase Method

- **PHASE 1:** Add an additional variable to the equation with the excess variable and minimize that variable:

$$\begin{aligned} \text{minimize} \quad & z' = a_2 \\ \text{subject to} \quad & x_1 + x_2 + s_1 = 10 \\ & -x_1 + x_2 - e_2 + a_2 = 2 \\ & x_1, x_2, s_1, e_2, a_2 \geq 0 \end{aligned}$$

- In tableau form:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	0	0	0	0	1	0
s_1	1	1	1	0	0	10
a_2	−1	1	0	−1	1	2

- This is not valid for optimality, because the top row must be expressed only in terms of non-basic variables. We can use row operations to make the a_2 term equal to zero:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	1	−1	0	1	0	−2
s_1	1	1	1	0	0	10
a_2	−1	1	0	−1	1	2

Two Phase (con't)

- After one iteration:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	0	0	0	0	1	0
s_1	-2	0	-1	-1	1	-8
x_2	-1	1	0	-1	1	2

Optimal value: $z' = 0,\ x_1 = 0,\ x_2 = 2, s_1 = 8, e_2 = 0, a_2 = 0$

- This only gives us a basis for our "real" optimization problem, so now ***Phase 2*** will take this basis for the original problem. We need to re-write the constraints to do this, since they can only have one basic variable in each one. The second one is fine, the first must have x_2 replaced:

$$\begin{aligned} \text{minimize} \quad & z = -2x_1 - x_2 \\ \text{subject to} \quad & x_1 + x_2 + s_1 = 10 \\ & -x_1 + x_2 - e_2 = 2 \\ & x_1, x_2, s_1, e_2 \geq 0 \end{aligned}$$



$$\begin{aligned} \text{minimize} \quad & z = -2x_1 - x_2 \\ \text{subject to} \quad & 2x_1 + s_1 + e_2 = 8 \\ & -x_1 + x_2 - e_2 = 2 \\ & x_1, x_2, s_1, e_2 \geq 0 \end{aligned}$$

NOTE: This can be read from the ending Tableau from Phase 1.

Two Phase (con't)

- Phase 2 Tableau:

variable	x_1	x_2	s_1	e_2	rhs
$-z$	-2	-1	0	0	0
s_1	2	0	1	1	8
x_2	-1	1	0	-1	2

- Make basic variables have zero value in the objective row:

variable	x_1	x_2	s_1	e_2	rhs
$-z$	-3	0	0	-1	2
s_1	2	0	1	1	8
x_2	-1	1	0	-1	2

Two Phase (con't)

- After one iteration:

variable	x_1	x_2	s_1	e_2	rhs
$-z$	0	0	$3/2$	$1/2$	14
x_1	1	0	$1/2$	$1/2$	4
x_2	0	1	$1/2$	$-1/2$	6

Optimal value: $z = -14,\ x_1 = 4,\ x_2 = 6, s_1 = 0, e_2 = 0$

Part 3:Big-M Method

Big-M Method of Artificial Variables

- We can combine the two phases of the previous approach by multiplying the artificial variables by a large value, called M, and then including the artificial variables in the original objective:

$$
\begin{aligned}
\text{minimize} \quad & z = -2x_1 - x_2 \\
\text{subject to} \quad & x_1 + x_2 + s_1 = 10 \\
& -x_1 + x_2 - e_2 = 2 \\
& x_1, x_2, s_1, e_2 \geq 0
\end{aligned}
$$



$$
\begin{aligned}
\text{minimize} \quad & z' = -2x_1 - x_2 + Ma_2 \\
\text{subject to} \quad & x_1 + x_2 + s_1 = 10 \\
& -x_1 + x_2 - e_2 + a_2 = 2 \\
& x_1, x_2, s_1, e_2, a_2 \geq 0
\end{aligned}
$$

Big-M Method on our Example

- Initial Tableau:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	-2	-1	0	0	M	0
s_1	1	1	1	0	0	10
a_2	-1	1	0	-1	1	2

There is a problem with this initial tableau – what is it?

- Re-write top row in terms of non-basic variables only:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	$M-2$	$-M-1$	0	M	0	$-2M$
s_1	1	1	1	0	0	10
a_2	-1	1	0	-1	1	2

Big-M Example (con't)

- After one iteration:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	-3	0	0	-1	$M+1$	2
s_1	2	0	1	1	-1	8
x_2	-1	1	0	-1	1	2

- After two iterations:

variable	x_1	x_2	s_1	e_2	a_2	rhs
$-z'$	0	0	$3/2$	$1/2$	$M-1/2$	14
x_1	1	0	$1/2$	$1/2$	$-1/2$	4
x_2	0	1	$1/2$	$-1/2$	$1/2$	6

Optimal value: $z = -14,\ x_1 = 4,\ x_2 = 6, s_1 = 0, e_2 = 0, a_2 = 0$

Part 4: Dual Simplex Problems

Example: Cabinet manufacturing

- Maximize weekly revenue of a woodworking factory
- Various orders will take various amounts of materials and workers

Product	Wood Required (bf)	Labor Required (hr)	Revenue ($)
Bookshelf	10	2	100
Cabinet w/ doors	12	4	150
Cabinet w/ drawers	25	8	200
Custom	20	12	400

- We have 5000 bf of wood available and 1500 labor hours
- What is the best combination to maximize $'s?

Dual Problems

- The simplicity of the form of the Linear Programming Problem leads to an equivalent problem (called the Dual Problem) where constraints and variables are flipped around
  - Why might this be useful?
  - What does it mean in the physical/practical problem?

- Example from first class:

Product	Wood Required (bf)	Labor Required (hr)	Revenue ($)
Bookshelf	10	2	100
Cabinet w/ doors	12	4	150
Cabinet w/ drawers	25	8	200
Custom	20	12	400

Given 5000 bf of wood available and 1500 labor hours, we wanted to maximize revenue ($$)

Linear Program:

$$\begin{aligned} \text{maximize} \quad & z = 100x_1 + 150x_2 + 200x_3 + 400x_4 \\ \text{subject to} \quad & 10x_1 + 12x_2 + 25x_3 + 20x_4 \leq 5000 \\ & 2x_1 + 4x_2 + 8x_3 + 12x_4 \leq 1500 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{minimize} \quad & z = -100x_1 - 150x_2 - 200x_3 - 400x_4 \\ \text{subject to} \quad & 10x_1 + 12x_2 + 25x_3 + 20x_4 + s_1 = 5000 \\ & 2x_1 + 4x_2 + 8x_3 + 12x_4 + s_2 = 1500 \\ & x_1, x_2, x_3, x_4, s_1, s_2 \geq 0 \end{aligned}$$

Duel Problems

- For a Linear Problem in canonical form, there is an equivalent form called the Dual Form given as follows:

Canonical Form

$$\begin{aligned} \text{minimize} \quad & z = c^T x \\ \text{subject to} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

Dual Form

$$\begin{aligned} \text{maximize} \quad & w = b^T y \\ \text{subject to} \quad & A^T y \leq c \\ & y \geq 0 \end{aligned}$$

- What does this mean for the woodshop problem?

$$\begin{aligned} \text{minimize} \quad & z = -100x_1 - 150x_2 - 200x_3 - 400x_4 \\ \text{subject to} \quad & -10x_1 - 12x_2 - 25x_3 - 20x_4 \geq -5000 \\ & -2x_1 - 4x_2 - 8x_3 - 12x_4 \geq -1500 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Product	Wood Required (bf)	Labor Required (hr)	Revenue ($)
Bookshelf	10	2	100
Cabinet w/ doors	12	4	150
Cabinet w/ drawers	25	8	200
Custom	20	12	400

Given 5000 bf of wood available and 1500 labor hours, we wanted to maximize revenue ($$)

$$\begin{aligned} \text{maximize} \quad & w = -5000y_1 - 1500y_2 \\ \text{subject to} \quad & -10\,y_1 - 2y_2 \leq -100 \\ & -12y_1 - 4y_2 \leq -150 \\ & -25y_1 - 8y_2 \leq -200 \\ & -20y_1 - 12y_2 \leq -400 \\ & y_1, y_2 \geq 0 \end{aligned}$$

The dual variables y_1, y_2 represent the additional profit each additional unit of wood and labor would provide – they are the marginal gains!
(to see this, re-write the dual form as a minimization)

Connections between Primal and Dual

- An LP in canonical form is called the ***primal problem*** and the LP in dual form is the corresponding ***dual problem***

PRIMAL PROBLEM

$$\begin{aligned} \text{minimize} \quad & z = c^T x \\ \text{subject to} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

DUAL PROBLEM

$$\begin{aligned} \text{maximize} \quad & w = b^T y \\ \text{subject to} \quad & A^T y \leq c \\ & y \geq 0 \end{aligned}$$

- Note that if a solution is optimal for both primal and dual then it solves both problems – and the objective values are the same.

COMPLEMENTARY SLACKNESS THEOREM: If x is optimal for the primal problem and y is optimal for the dual problem, then $x^T(c - A^T y) = 0$. Also, if (i) x is feasible for the primal problem, (ii) y is feasible for the dual problem, and (iii) $x^T(c - A^T y) = 0$, then x and y are optimal for the primal and dual problems.

- What does complementary slackness mean? This means that for any bound constraint that is not binding for the primal (i.e. where the equality part does not hold for $x \geq 0$), the corresponding regular constraint in the dual must be binding (i.e. the equality part holds in $A^T y \leq c$). The same goes for bound constraints for the dual influencing the regular constraints in the primal.
- These properties are useful to check solutions – especially when a computer solution doesn't make sense.

Solving Dual Problems

- Dual problem form comes from the canonical form (the primal problem)
- To move from standard form to canonical form we need to make the $=$ constraints become $\geq$ constraints
- Approach:
  - Make the objective in standard form (convert to minimization and remove constants)
  - Add extra variables and/or constraints to fix the bounds on the variables
  - DO NOT add any slack or excess variables to regular constraints (ok if you do, just makes more work)
  - For any $\leq$ constraints, multiply by minus one (recall it's ok in canonical form to have $b < 0$)
  - For any natural equality constraints, we replace with two constraints:
    - $a^{\mathrm{T}}x = b$ ➔ $a^{\mathrm{T}}x \leq b$ and $a^{\mathrm{T}}x \geq b$ ➔ $-a^{\mathrm{T}}x \geq -b$ and $a^{\mathrm{T}}x \geq b$
- This is typically fewer variables than standard form (maybe slightly more equations)
- Now form the dual problem:
  - Make the objective in standard form (convert to minimization and remove constants)
  - new variables y
  - transpose the constraint matrix A
  - convert the objective coefficients c^{T} into the constraint constants c
  - convert the constraint constants b into the objective coefficients b^{T}

White Board Examples

When to do a Dual versus a Primal?

- Duals provide additional features not found in the primal:
    1. If the primal gives a non-standard answer (such as unbounded) the dual can be used to see if that makes sense
    2. Sometimes with lots of variables and few constraints, the dual is easier to solve than the primal. Why?
    3. Additionally, duals can be used to formulate valuations

- Duals can also be used to find feasible solutions for the primal basis when not easy to find (this is actually what most commercial linear programming solvers do instead of adding artificial variables like in two-phase or big-M). This is something called the dual simplex, it is mentioned in most texts and is a simple extension of the simplex tableau method.

Dual problems for Valuation

Example: Recall the woodshop problem. What if you own the woodshop and I want to buy the woodshop from you at the cheapest reasonable price? How do I find a fair value to make an offer?

Product	Wood Required (bf)	Labor Required (hr)	Revenue ($)
Bookshelf	10	2	100
Cabinet w/ doors	12	4	150
Cabinet w/ drawers	25	8	200
Custom	20	12	400

Given 5000 bf of wood available and 1500 labor hours, we wanted to maximize revenue ($$)

I could ask myself the question of how I can minimize the amount paid for assets (which are labor and materials). Let y_1 be the value I will pay per unit (bf) of wood and y_2 be the value per unit (hour) of labor.

Then I want my objective to be: $\text{minimize } w = 5000\, y_1 + 1500\, y_2$

Furthermore I want to make this fair to you as the current owner so that would mean getting at least the revenue that you get from current operations.

So $10y_1 + 2y_2 \geq 100$ is a constraint that the price you get for wood and labor is at least as good as the revenue that you would get from using wood and labor to make bookshelves.

Continuing on with each type of product, we get…

Dual problem for woodshop

$$
\begin{aligned}
\text{minimize} \quad & w = 5000y_1 + 1500y_2 \\
\text{subject to} \quad & 10\,y_1 + 2y_2 \geq 100 \\
& 12y_1 + 4y_2 \geq 150 \\
& 25y_1 + 8y_2 \geq 200 \\
& 20y_1 + 12y_2 \geq 400 \\
& y_1, y_2 \geq 0
\end{aligned}
$$

- Converting to maximization and making constraints be $\leq$ leads to:

$$
\begin{aligned}
\text{maximize} \quad & w = -5000y_1 - 1500y_2 \\
\text{subject to} \quad & -10\,y_1 - 2y_2 \leq -100 \\
& -12y_1 - 4y_2 \leq -150 \\
& -25y_1 - 8y_2 \leq -200 \\
& -20y_1 - 12y_2 \leq -400 \\
& y_1, y_2 \geq 0
\end{aligned}
$$

Dual form of woodshop problem

- For the optimal value, y_1 and y_2 are the fair valuations for the resources of the woodshop (wood and labor)
- Because of this, economists refer to dual variables as imputed costs

Solution of woodshop problem

- Primal solution:

$$(x_1, x_2, x_3, x_{4)} = (37.5, 0, 0, 62.5)$$
$$z = -62500$$

This means making 37.5 bookshelves and 62.5 custom builds for a revenue of $62,500

- Dual solution:

$$y_1, y_2 \ = (5, 25)$$
$$w = -62500$$

This means I should be willing to pay up to $5/bf for material and $25/hour for labor

- This also tells both of us the marginal gains we get from both labor and materials. If we are paying more than $25/hour for labor it is too much relative to the material costs and revenues. Also if we are paying more than $5/bf for materials you are going to need to raise prices of your goods.
- Could you have figured out the dual from the primal? Or vice-versa?

Complementary Slackness: $-10\,y_1 - 2y_2 = -100$ and $-20y_1 - 12y_2 = -400$ ➔ $(y_1, y_2) = (5, 25)$

Practice Problems for Midterm

Advanced Optimization and Operations Research Textbook

- Simplex Method: Section 3.13 (pg 76)
- Revised Simplex: Section 4.9 (pg 107)
- Dual Simplex: Section 5.6 (pg 124)